



Delivery Operations Analytics

Diagnosing why delivery times vary across 45,000+ quick-commerce orders and turning it into three specific, actionable recommendations.

Excel (Power Query)

SQL

Microsoft Fabric

Power BI

Prepared for: Analytics Leadership Review

THE BUSINESS PROBLEM

Nobody knew why delivery times varied or how to find out fast

MealDash's delivery times swing widely: some orders arrive in 10 minutes, others take almost an hour. Diagnosing a single slow week meant a data analyst manually digging through raw records for a day or two, with no standing answer to the question ops kept asking:

“Why was it slow and what can we actually do about it?”

This project answers 10 specific business questions — moving from the overall scale of the problem, to the levers ops can control (traffic, distance, staffing, bundling) versus the ones they can't (weather), and finally to whether performance is trending better or worse over time.

10 – 54 min

Delivery time range observed

1–2 Days

To manually diagnose a slow week — before this project

Minutes

To answer the same question — after this project

A real, messy, 45,000-order dataset nothing synthetic

What's in the data

- Delivery timestamps & GPS coordinates
- Weather conditions & road traffic density
- Delivery agent ratings & vehicle type
- Festival-day flags
- ≈45,000 real orders — Zomato Delivery
- Operations dataset (Kaggle)

✓ In Scope

- Traffic
- Weather
- Distance
- Vehicle type
- Agent performance
- Order bundling
- Festival effects

✗ Out of Scope

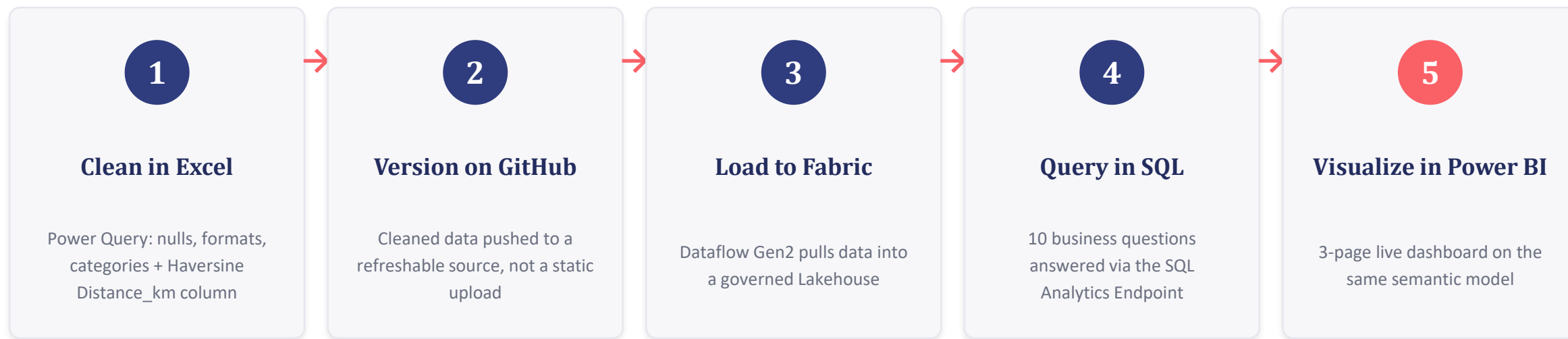
- Pricing
- Restaurant menu data
- Customer app experience

Left out on purpose — pulling them in would dilute a focused delivery-time analysis.

Note: no “promised delivery time” field exists, so SLA breach rate is benchmarked against segment averages instead of a formal target.

APPROACH

A governed, repeatable pipeline not a one-off manual pull



Choose from predefined task flows or add a task to build one
Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

[Select a predefined task flow](#) [Add a task](#) [Import a task flow](#)

	Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorse...	Sensitivity	Included in app
	MealDash_Dashboard		Report	—	MealDash	8/21/2026, 1:10...	—	—	—	☐ No
	MealDash_Dataflow	✓	Dataflow Ge...	—	SEEMA KU...	8/21/2026, 11:2...	—	—	—	
	MealDash_Lakehouse		Lakehouse	—	SEEMA KU...	—	—	—	—	
	MealDash_Lakehouse		SQL analytic...	—	SEEMA KU...	—	—	—	—	
	MealDash_Schematic		Semantic m...	—	MealDash	8/21/2026, 1:1...	N/A	—	—	

Fabric workspace: Lakehouse, Dataflow Gen2 & SQL Analytics Endpoint under one governed roof

Each tool earns its place in the pipeline

1

Excel · Power Query

First cleaning pass: nulls, formats, categorical standardization, and a Haversine-based Distance_km column.

2

SQL · Fabric Analytics Endpoint

Answered all 10 business questions using CTEs, CASE-based bucketing, and window functions (PERCENTILE_CONT, rolling averages).

3

Microsoft Fabric · Lakehouse + Dataflow Gen2

A repeatable, governed pipeline pulling cleaned data from GitHub into a Lakehouse instead of a one-off manual upload.

4

Power BI

A 3-page live, interactive dashboard with DAX measures, built on the same governed semantic model as the SQL layer.

Baseline, city, traffic & weather

26 min

Overall avg (10–54 min range)

27 min

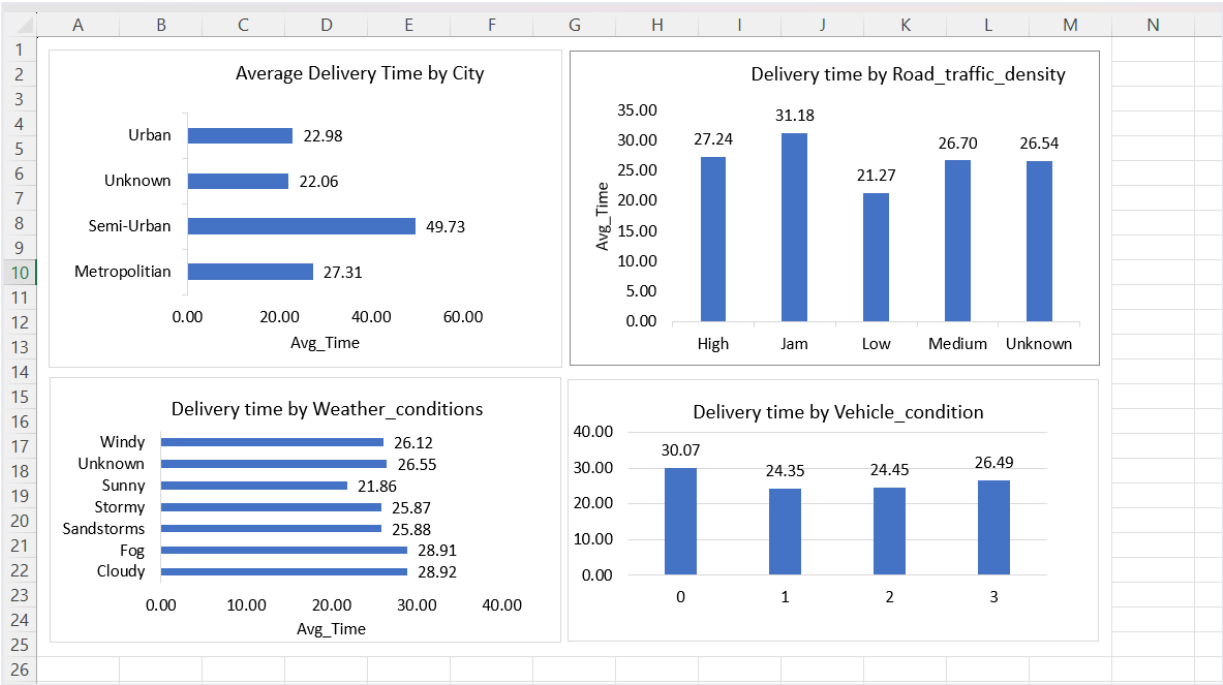
Metropolitan — slowest by volume
(34,000+ orders)

+10 min

Jam traffic vs. Low traffic

28 min

Fog & Cloudy — slowest weather
(not storms)



Distance, vehicle, agent rating & bundling



Distance

Crossing the 5 km mark is the real cost — time plateaus after that, it isn't proportional to distance.

Vehicle type

Minor effect (~3 min) between fastest and slowest vehicle.

Agent rating

4.5+ rated agents deliver 10–13 min faster than every lower bucket — the clearest pattern in the dataset.

Bundled orders

2–3 bundled orders more than doubles delivery time (22 → 47 min), and the cost accelerates with each extra stop.

Festival impact & the trend over time

75–80%

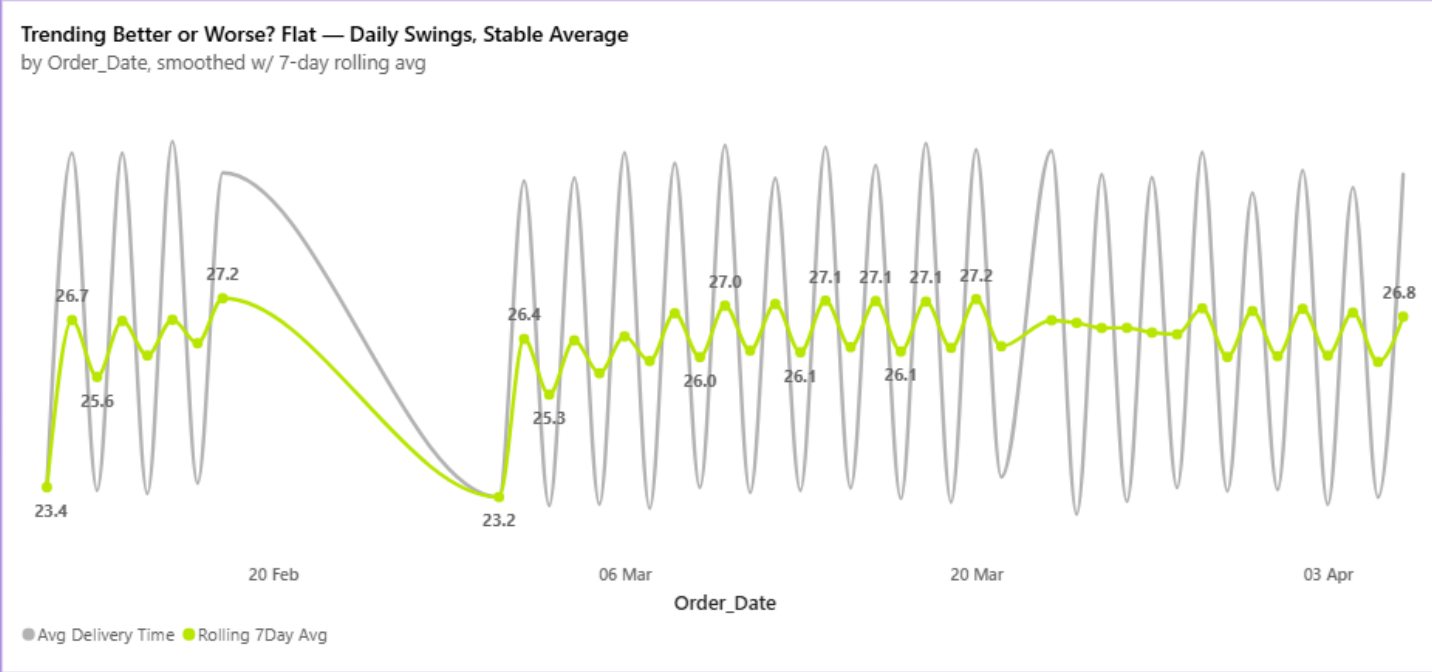
Festival days are slower than normal days — the largest single effect found across all 10 questions

Flat

7-day rolling average holds steady around 25–26 min — no long-term drift; patterns are structural

MealDash

Is delivery time trending better or worse over time?



Daily averages fluctuate between 23–30 minutes, but the 7-day rolling average stays flat around 25–26 minutes across the full period no long-term drift. This confirms the patterns found in Q2–Q9 are structural, not caused by delivery performance getting better or worse over time.

A 3-page Power BI dashboard, built for self-serve exploration



Page 1 — Executive Overview

Total orders, avg delivery time, festival impact %, and slow-delivery rate at a glance.

Page 2 — Delivery Drivers Deep Dive

Distance, rating, vehicle condition and bundling, all slicer-driven.

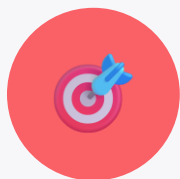
Page 3 — Trend Over Time

Daily averages smoothed with a 7-day rolling average to separate noise from signal.

[Open the Live Interactive Dashboard](#)

RECOMMENDATIONS

Three findings ops can act on this week



Pre-plan festival-day staffing

Festival deliveries take 75–80% longer — the single largest effect found. Low volume (896 orders) but high impact makes this a high-ROI, low-effort fix: targeted staffing on known festival dates, not a company-wide change.



Cap bundled deliveries at 1 extra order

2–3 bundled orders more than doubles delivery time (22 → 47 min), and the cost accelerates non-linearly. Capping at 1 extra order for time-sensitive deliveries captures most of the efficiency without the steep penalty.



Use agent ratings for order routing

4.5+ rated agents deliver in 24 min vs. 34–37 min for every lower bucket — a 10+ minute gap. A strong, easy-to-access signal for prioritizing time-sensitive orders, even though causal direction isn't fully resolved.

Traffic and weather matter too — but they're conditions ops can't directly control. These three are things ops can act on now.

WHAT'S NEXT

Future Work

- 1 Validate the festival-day effect against a full year of festival dates
- 2 Replace straight-line (Haversine) distance with actual road-route distance
- 3 Investigate the direction of causation between agent rating and delivery speed
- 4 Add a live data connection to support ongoing monitoring, not one-time diagnosis

Thank you

Open to opportunities, collaborations, and conversations around data analytics.

github.com/seema-kri/MealDash